FORCE AND LAWS OF MOTION

Class 9 | Science | Chapter 9

FORCE

A push or a pull on an object that can change its state of rest/motion, speed, direction, or shape.

Balanced forces: no change in state of motion (e.g. tug of war at a draw). Unbalanced forces: cause a change in motion.

NEWTON'S FIRST LAW OF MOTION

An object remains in a state of rest or uniform motion in a straight line unless acted upon by an unbalanced (external) force.

Also called the Law of Inertia. Inertia = tendency of an object to resist a change in its state of rest or motion.

TYPES OF INERTIA

Type	Example
Inertia of rest	A person jerks backward when a bus suddenly starts
Inertia of motion	A person falls forward when a moving bus suddenly stops
Inertia of direction	Mud flies off a rotating wheel tangentially

Mass is a measure of inertia -- more mass means more inertia.

NEWTON'S SECOND LAW

Momentum & Force Formula

MOMENTUM

Momentum (p) = mass x velocity (p = mv). SI unit: kg m/s. It is a vector quantity.

NEWTON'S SECOND LAW OF MOTION

The rate of change of momentum of an object is proportional to the applied unbalanced force, in the direction of the force.

F = m x a

Where F = force, m = mass, a = acceleration. SI unit of force: newton (N) = 1 kg m/s^2.

Derivation: F = (mv - mu)/t = m(v-u)/t = m x a

APPLICATIONS

- A karate expert can break a slab with a swift blow -- large force applied in a very short time.
- Cricketers pull their hands back while catching a fast ball -- increases the time of impact, reducing the force.
- Long jump pits have sand -- increases time of stopping, reducing force of impact on the athlete.

NEWTON'S THIRD LAW

Action-Reaction & Conservation of Momentum

NEWTON'S THIRD LAW OF MOTION

For every action, there is an equal and opposite reaction -- and they act on two different objects.

Ex: A gun recoils backward when a bullet is fired forward. Walking -- foot pushes ground back, ground pushes foot forward.

LAW OF CONSERVATION OF MOMENTUM

In the absence of an external unbalanced force, the total momentum of a system of objects remains constant (unchanged).

For two objects colliding: $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$

Ex: Recoil of a gun -- momentum of bullet forward = momentum of gun backward, so total momentum stays zero.

SUMMARY & REVISION

Key Laws + Quick Recap

CHAPTER SUMMARY -- KEY TERMS

- **Inertia** -> resistance to change in state of rest/motion
- **1st Law** -> object stays at rest/uniform motion unless an unbalanced force acts
- **Momentum** -> mass x velocity
- **2nd Law** -> F = ma (force = rate of change of momentum)
- **3rd Law** -> every action has an equal & opposite reaction
- **Conservation of momentum** -> total momentum constant with no external force

SOLVED EXAMPLE

Q: A force of 10 N acts on a 2 kg object. Find its acceleration.

A: Using F = ma: a = F/m = 10/2 = 5 m/s^2.

ONE-LINE RECAP FOR REVISION

1st Law: inertia -- objects resist changes in motion.

2nd Law: F = ma -- links force, mass and acceleration.

3rd Law: action-reaction pairs; momentum is conserved.

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